CASEFILE · STAGE 6 DETECTIVE FOLLOW-UP

10,000 burglaries.
A few hundred burglars.

Cascade City logged 10,000 burglaries and cleared 1,300 (13%). Casefile connects the cases so detectives can close them.

AI suggests

Agents read reports, hypothesize series, draft packets.

A rules engine verifies

Could one person be at both scenes? Same MO? Serial match? Yes/no with reasons.

People decide

Every assignment, alert and victim message needs a click, recorded in a hash-chained audit log.

Cases, not people: we never score a person or a neighbourhood. Prototype runs on synthetic data plus public Seattle geography.

THE BOTTLENECK

Connecting cases, not collecting them

Official baseline: **$305 per case, $3.05M a year.**

STEP	MINUTES	WHO
Intake	15	Officer
Report	60	Officer
Video review	120	Officer
Evidence	45	Officer
Detective follow-up	**90**	**Detective: $1.05M, the largest cost**

Where an hour is least valuable

A detective re-reading a no-lead report and cross-referencing by hand.

- Clearance counts offenses: one arrest can clear every linked burglary.
- A few repeat offenders commit a large share of burglaries; most cases never get linked.
- Domain expert at kickoff: "We know who the bad guys are. We just don't have enough resources."

Sources: case brief (baseline); Eck, PERF 1983 (most burglary investigations suspended early for lack of leads); FBI UCR clearance rules.

WHAT CHANGES ON MONDAY MORNING

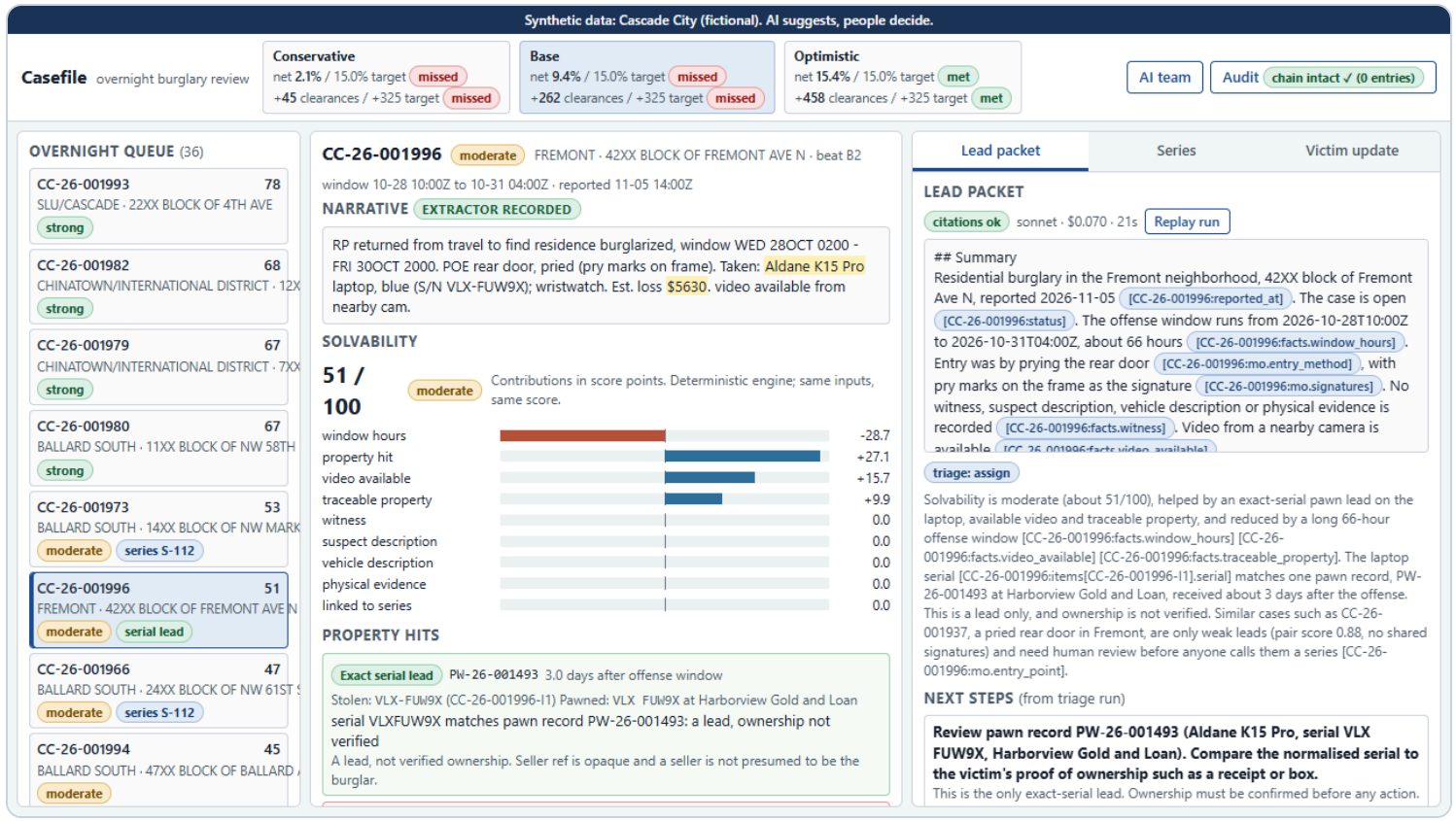
One screen for the sergeant and the detective

	TODAY	WITH CASEFILE
Sergeant	Reads 36 overnight reports one by one	Opens a queue already structured, scored and cross-referenced; screens out no-lead cases with a signed packet
Detective	Hunts through RMS and pawn records	Gets a lead packet: linked cases, pawn hits, a next-steps checklist to accept or reject with a reason
Victim	Hears nothing for weeks	Gets a plain-language update in their language, approved by staff

> Stage 7 property matching feeds Stage 6: pawn serial hits are leads, never proof of ownership or guilt (pawn records are furnished to police under RCW 19.60.040).

DEMO

Live from the prototype



75-second walkthrough, recorded runs of real Claude agents replayed offline:

- Overnight queue of 36 reports
- Serial hit vs. look-alike rejected
- Series proposed, then one case ruled out by the engine
- Packet with citations
- Human approval, audit log

ARCHITECTURE

AI proposes. The engine verifies. A detective decides.

Connectors

MCP server (Rust): records, pawn feed, **verify_series**, solvability, citation checker, approval queue, cost model

Agents

Extractor (Haiku) · Triage (Sonnet) · Series Hunter (Sonnet) · Skeptic (Opus) · Packet Writer (Sonnet) · Victim Liaison (Haiku)

Teams

Intake & Triage (per report) · Series Hunting (hunter ⇄ skeptic ⇄ engine, nightly) · Victim Comms (on status change)

Channels

Sergeant console · detective packet · victim message draft (any language, staff-approved). Not built: patrol hands-free brief.

> Approvals never block: requests queue with a ticket and escalate after 120 min, so human review waits hours, not days.

Single LLM vs. proposer + verifier (our eval)

CASES	LLM ONLY F1	CASEFILE F1	DECOY PAIRS ACCEPTED BY LLM
20	100%	**80%**	0
60	67%	**69%**	6
150	86%	**86%**	9

Planted series with decoys, synthetic ground truth, one run per size. The LLM matches us on F1 at small scale; what it lacks is the hard check: it accepted decoy pairs that one person cannot have committed, and 150 cases already cost about 53k input tokens (2,000 would be about 0.7M, over one context window).

Full 2,000-case run

Series pairs: precision 66%, recall 100%, 38/40 planted series recovered. Decoys rejected: 10/10. Exact-serial leads: 100/100 correct. Extractor (12 recorded runs): entry, premise, weekday 100%; time band only 58% (wide windows are ambiguous; a human confirms).

COST-BENEFIT ON THE OFFICIAL BASELINE

Base case: 9.4% net, short of the 15% target

$285k

net per year, 9.4% of $3.05M (target 15% = $457.5k)

Detective + officer capacity freed	$428k
Run cost	-$62k
Funded follow-up of new leads	-$76k
Dismissing false leads	-$4k
One-time setup	$130k
Year one (ramped)	$6k

SCENARIO	NET %	+CLEARANCES	PAYBACK
Conservative	2.1%	+45	4.1 y
Base	9.4%	+262	1.0 y
Optimistic	15.4%	+458	0.4 y

> We probably miss 15%. Only the Optimistic bracket reaches it. Dollars are freed capacity (about 5,100 detective hours), not cash; freed hours fund follow-up, no layoffs. Cash savings need avoidable overtime or unfilled posts; finance must confirm.

The team research bot's more cautious bracket (3.5% net) sits between our Conservative and Base. The pilot decides.

CLEARANCE LOGIC

How 13% becomes 15.6%: new leads, worked by detectives

STEP (BASE)	CASES/YR
Open cases linked to an arrest (1,300 cleared × 30% hit × 1.5 per hit × 85% adoption)	497
Pawn/serial hits (10,000 × 35% traceable × 10% resurface × 85% × (1 − 20% already linked))	238
Moderate cases newly worked (120 × 85% adoption)	102
New leads → 65% completed → 50% clear, minus 10 lost to screening	**837 → +262**

> Base gives 15.6%, not the 16.25% target (+325). Only Optimistic (+458) clears it. The 50% conversion is unbenchmarked (today's clearance is 13%); at the conservative 35% the result is +45. Every factor is an assumption the pilot measures.

Follow-up hours are a cost, not a saving. False links cost detective time (184 false leads dismissed at 20 min each, 80% precision assumed; our eval measures 66% pair precision on synthetic data, so the real number is a pilot question).

PILOT

Prove it on one precinct before scaling

- **Days 0–30:** integrate the RMS export and pawn feed in a CJIS-approved environment; train 1 sergeant and 6 detectives; replay 6 months of closed cases to measure link precision.
- **Days 31–90:** randomized pilot: half of new burglaries get Casefile packets, half business as usual.
- **Months 4–12:** expand only if the gates pass; quarterly accuracy and equity audits.

> **The one metric: cleared burglaries per detective-hour, pilot vs. control.** It captures both asks, savings and clearance, at once.

Gates: link precision on replay meets the bar we set with the department; zero un-reviewed actions; no drop in clearance on screened cases.

ETHICS, TRUST AND THE ASK

Cases, not people. Humans decide. Everything is auditable.

- **Bias:** links cases, never scores people or neighbourhoods; no predictive patrol; equity monitor on screening rates.
- **Privacy:** CJIS-approved hosting; only data the department lawfully holds; no facial recognition.
- **False positives:** skeptic agent plus rules engine rejected 10/10 decoy series; a serial match is a lead, not ownership.
- **Human in the loop:** no send, assign or alert action exists without a click; hash-chained audit log; replays give identical output.

Measured inference cost (recorded runs)

AGENT	MODEL	$/RUN	N
extractor	haiku	$0.002	12
triage	sonnet	$0.036	4
series_hunter	sonnet	$0.078	1
skeptic	opus	$0.176	1
packet_writer	sonnet	$0.063	2
victim_liaison	haiku	$0.002	1
series_llm_only	opus	$0.266	3

Right-sized models: the cheapest adequate model per task. Per-case chain (extractor + triage + packet) about $0.10, so roughly $1.0k a year at 10,000 cases, against the $12k/yr inference line in the Base budget.

Ask: approve a 90-day, one-precinct pilot; scale only if the gates pass. Full program: $130k setup, $62k/yr to run.